

Assignment Problem

Hungarian Method

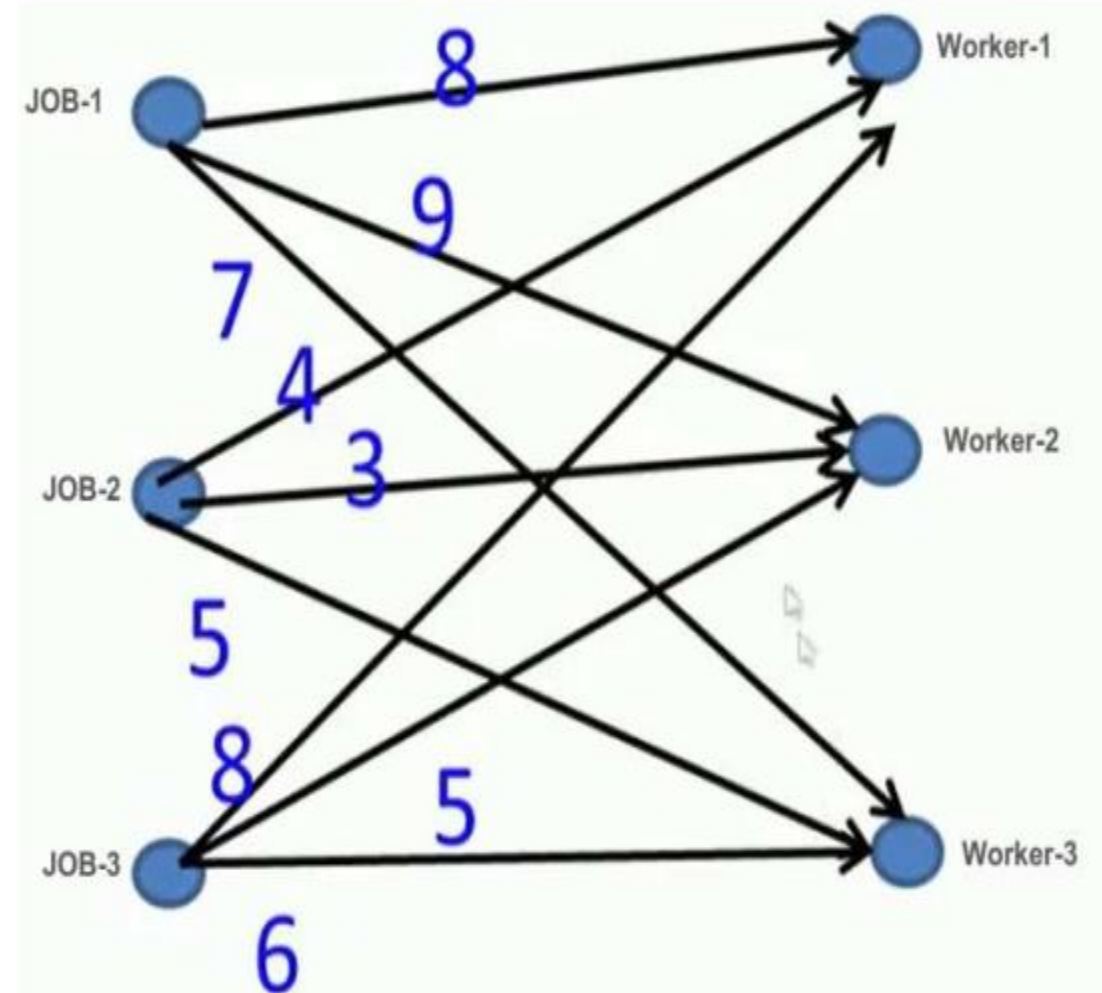
Assignment Problem

- ✓ In many business situations, management needs to assign - personnel to jobs, jobs to machines, machines to job locations, salespersons to territories or project-teams to projects.
- ✓ Consider the situation of assigning n jobs to n workers.
- ✓ When a Job i ($=1,2,\dots,n$) is assigned to worker j ($=1,2, \dots,n$) that incurs a cost C_{ij}
- ✓ The objective is to assign the jobs to workers at the least possible total cost.
- ✓ The cost associated is C_{ij} when Job i is assigned Worker j .

$$C_{11}=8, C_{12}=9, C_{13}=7,$$

$$C_{21}=4, C_{22}=3, C_{23}=5,$$

$$C_{31}=8, C_{32}=5, C_{33}=6.$$



Assignment Problem

- ✓ This situation is a special case of the Transportation Model And it is known as the *assignment problem*.
- ✓ Here, jobs represent “sources” and machines represent “destinations.”
- ✓ The supply available at each source is 1 unit And demand at each destination is 1 unit.

	Machine					Source
		1	2		n	
Job	1	C11	C12		C1n	1
	2	C21	C22		C2n	1
	.	.	.		.	.
	.	.	.		.	.
	.	.	.		.	.
	n	Cn1	Cn2		Cnn	1
Destination		1	1		1	

Assignment Problem

Formulation

Let, $X_{ij} = 1$, if job i is assigned to worker j

$X_{ij} = 0$, otherwise

X_{11} ,

X_{12} ,

X_{13} ,

X_{21} ,

X_{22} ,

X_{23} ,

X_{31} ,

X_{32} ,

X_{33} .

Objective Function,

$$\text{Min } Z = C_{11}X_{11} + C_{12}X_{12} + C_{13}X_{13} + C_{21}X_{21} + C_{22}X_{22} + C_{23}X_{23} + C_{31}X_{31} + C_{32}X_{32} + C_{33}X_{33}$$

Each job goes to exactly one worker.

$$X_{11} + X_{21} + X_{31} = 1$$

$$X_{12} + X_{22} + X_{32} = 1$$

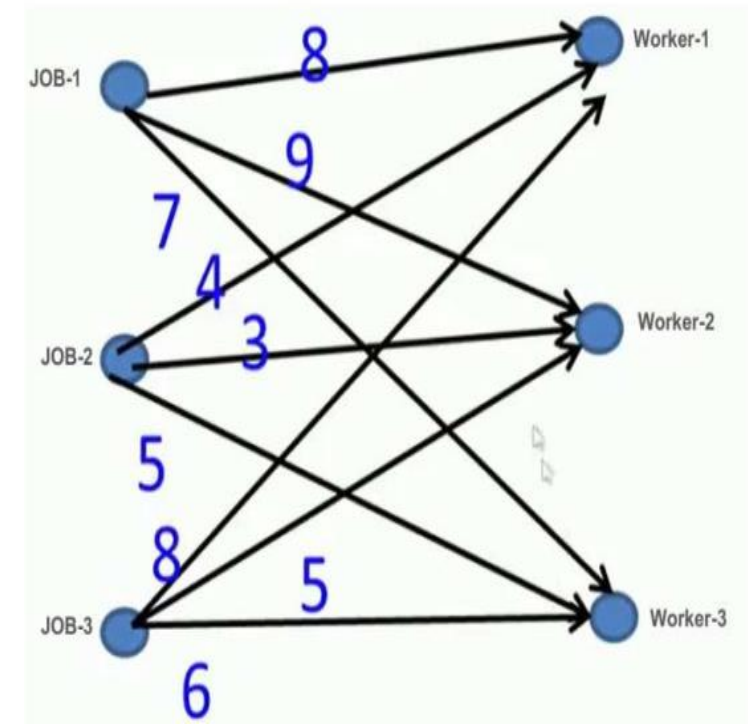
$$X_{13} + X_{23} + X_{33} = 1$$

Each worker gets only one job.

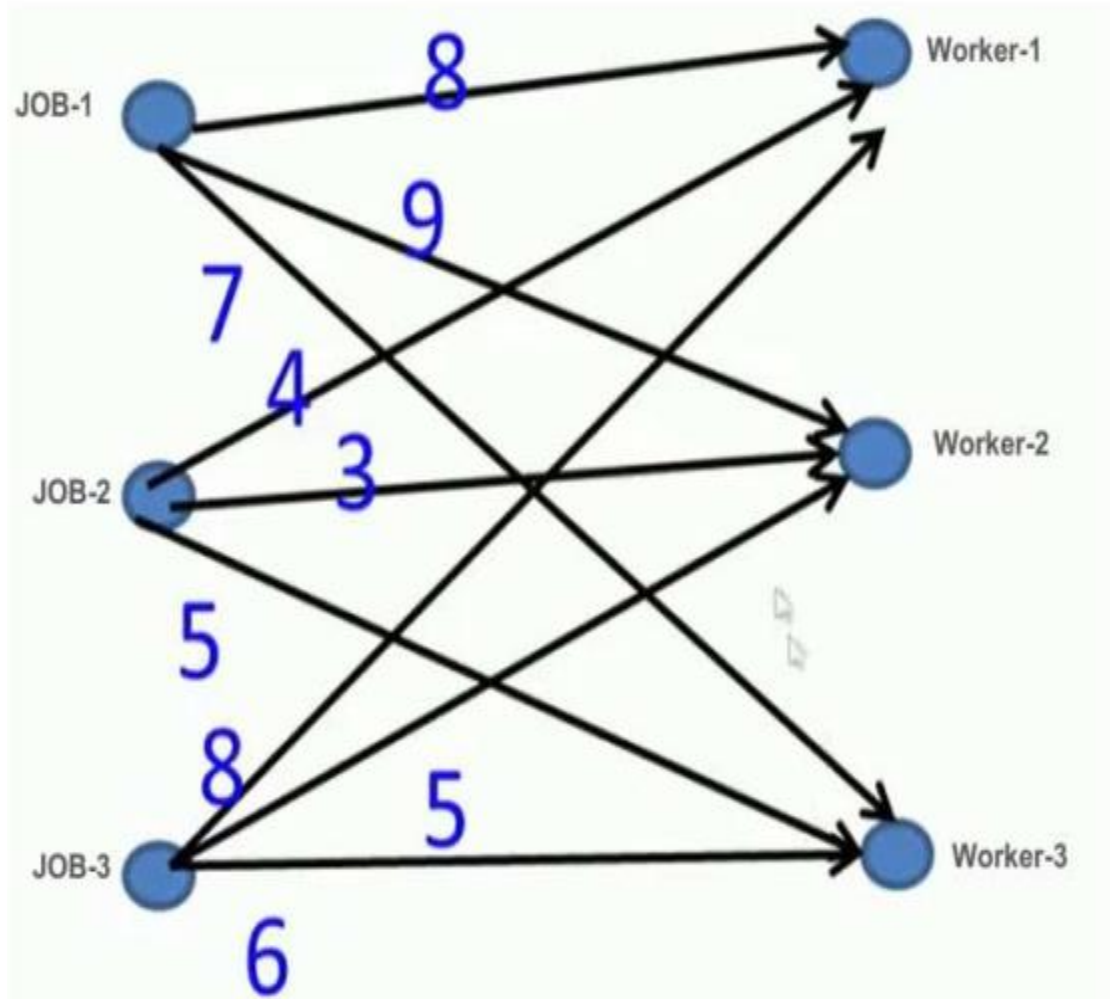
$$X_{11} + X_{12} + X_{13} = 1$$

$$X_{21} + X_{22} + X_{23} = 1$$

$$X_{31} + X_{32} + X_{33} = 1$$



Assignment Problem



	W1	W2	W3
J1	C_{11}	C_{12}	C_{13}
J2	C_{21}	C_{22}	C_{23}
J3	C_{31}	C_{32}	C_{33}

	W1	W2	W3
J1	8	9	7
J2	4	3	5
J3	8	5	6

Assignment Problem: Hungarian Algorithm

Example 1

Step 1: Select the smallest value in each row. Subtract this value from each value in that row

Example

	W1	W2	W3	W4
J1	8	6	4	8
J2	6	5	5	8
J3	9	10	11	12
J4	7	6	8	10

	W1	W2	W3	W4
J1	8	6	4	8
J2	6	5	5	8
J3	9	10	11	12
J4	7	6	8	10

	W1	W2	W3	W4
J1	4	2	0	4
J2	1	0	0	3
J3	0	1	2	3
J4	1	0	2	4

Assignment Problem: Hungarian Algorithm

Example 1

Step 2: Identify column-minimum for each column & Subtract column-minimum from each respective column's element

	W1	W2	W3	W4
J1	4	2	0	4
J2	1	0	0	3
J3	0	1	2	3
J4	1	0	2	4

	W1	W2	W3	W4
J1	4	2	0	4
J2	1	0	0	3
J3	0	1	2	3
J4	1	0	2	4

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

Assignment Problem: Hungarian Algorithm

Example 1

Step 3: Make Assignment at zero position row-wise

- ✓ First make assignment for that row which has only one zero

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

Assignment Problem: Hungarian Algorithm

Example 1

Step 4: Make Assignment at vacant/unoccupied zero position column-wise, if any column has.

- ✓ First make assignment for that column which has only one zero unoccupied (neither assigned nor crossed)

	W1	W2	W3	W4
J1	4	2	0	1
J2	1	0	0	0
J3	0	1	2	0
J4	1	0	2	1

Example

	W1	W2	W3	W4
J1	8	6	4	8
J2	6	5	5	8
J3	9	10	11	12
J4	7	6	8	10

Feasible solution: Every row has one assignment and every column has one assignment

$$TC = 4 + 8 + 9 + 6 = 27$$

Assignment Problem: Hungarian Algorithm

Example 2

Step 1: Row operations

	W1	W2	W3	W4
J1	4	2	0	5
J2	2	0	1	4
J3	0	1	2	3
J4	3	0	2	5

Step 2: Column Operations

	W1	W2	W3	W4
J1	4	2	0	2
J2	2	0	1	1
J3	0	1	2	0
J4	3	0	2	2

	W1	W2	W3	W4
J1	8	6	4	9
J2	7	5	6	9
J3	9	10	11	12
J4	9	6	8	11

Assignment Problem: Hungarian Algorithm

Example 2

Step 3: Assignment

No more assignment possible because there is no vacant zero (All zeroes are covered either by assignment or crossed)

Feasibility: Solution is not feasible

Assignment Problem: Hungarian Algorithm

Example 2

Step 4: Draw a minimum number of lines through some of the rows and columns, Such that all zero values are crossed out.

(Steps-first, draw lines through row/column which has a greater number of zeroes than other)

Assignment Problem: Hungarian Algorithm

Example 2

Step 5: Select the smallest uncrossed out element. This element is subtracted from every uncrossed element and added to every element at the intersection of two lines.

Assignment Problem: Hungarian Algorithm

Example 2

Step 6: Again, perform assignment row-wise & Column-wise

Assignment Problem: Hungarian Algorithm

Example 2

Step 7: If drawing lines do not provide a feasible solution, then we should perform the task of drawing lines one more time.

Actually, we should continue drawing lines until a feasible assignment is possible.

Assignment Problem: Hungarian Algorithm

Example 3: Multiple solution

	Team 1	Team 2	Team 3	Team 4
Job 1	8	6	4	7
Job 2	6	5	5	8
Job 3	9	10	11	12
Job 4	7	6	8	10

Assignment Problem: Hungarian Algorithm

Example 4:

	Machine 1	Machine 2	Machine 3	Machine 4
Job 1	1	4	6	3
Job 2	9	7	10	9
Job 3	4	5	11	7
Job 4	8	7	8	5

0	2	1	1
3	0	0	2
0	0	3	2
4	2	0	0

<u>1</u>	4	6	3
9	<u>7</u>	10	9
<u>4</u>	5	11	7
8	7	8	<u>5</u>

0	3	<u>5</u>	2
2	0	<u>3</u>	2
0	1	<u>7</u>	3
3	2	<u>3</u>	0

0	3	2	2
2	0	0	2
0	1	4	3
3	2	0	0

0	3	2	2
2	0	0	2
0	1	4	3
3	2	0	0



Assignment Problem: Hungarian Algorithm

Examples

4. (Multiple Optimal Solutions) solve the minimal assignment problem whose cost matrix is given below. Also, give all possible optimal solutions.

	1	2	3	4
I	2	3	4	5
II	4	5	6	7
III	7	8	9	8
IV	3	5	8	4

10. The assignment cost of assigning any one operator to any one machine is given in the following table.

		Operator			
		I	II	III	VI
Machine	A	10	5	13	15
	B	3	9	18	3
	C	10	7	3	2
	D	5	11	9	7

Find the optimal assignment.

Assignment Problem: Hungarian Algorithm

Examples

		Men				
		1	2	3	4	5
Job	I	20	15	18	20	25
	II	18	20	12	14	15
	III	21	23	25	27	25
	IV	17	18	21	23	20
	V	18	18	16	19	20

Assignment Problem: Hungarian Algorithm

Examples

Example 9.3 A company has 5 jobs to be done on five machines. Any job can be done on any machine. The cost of doing the jobs on different machines are given below. Assign the jobs for different machines so as to minimize the total cost.

Jobs	Machines				
	A	B	C	D	E
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20